ChatGPT Evaluation for QA Knowledge – Project Report

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1. INTRODUCTION

This project evaluates the performance and reliability of ChatGPT in the context of Quality Assurance (QA). A total of 25 QA-focused prompts were executed on ChatGPT to assess how well the model can support QA learning, testing concepts, scenario analysis, and reasoning.

The prompts were designed to mimic real QA tasks and were grouped into four categories:
Basic QA Concepts, Scenario-Based Questions, Edge Case Questions, and Consistency Checks. The project demonstrates an analytical approach to evaluating AI model quality and provides insight into ChatGPT's effectiveness as a QA knowledge assistant.

2. METHODOLOGY

Model Evaluated:

• ChatGPT (OpenAI)

Evaluation Steps:

1. Designed 25 QA prompts across four categories.
2. Executed each prompt directly on ChatGPT.
3. Captured responses and documented them in an evaluation sheet.
4. Scored responses based on:

• Accuracy
• Relevance
• Completeness
• Clarity
• Safety
• Consistency (for selected prompts)

5. Calculated an Overall Score and pass/fail status.

Scoring Scale (1–5):

• 1 = Poor, 5 = Excellent

Pass Criteria:

• Average Score ≥ 4.0

3. PROMPT CATEGORIES

• Basic QA Concepts (6 prompts)
• Scenario-Based QA Questions (8 prompts)
• Edge Case / Advanced QA Prompts (7 prompts)
• Consistency Questions (4 prompts)

Each category represents a different dimension of QA learning and testing behavior.

4. SUMMARY OF RESULTS

Average Scores:

• Accuracy: 4.64
• Relevance: 5.00
• Completeness: 4.00

• Clarity: 5.00
• Safety: 5.00
• Consistency: 5.00

Overall Performance:
ChatGPT performed strongly across QA theory, explanations, and clarity.
The model demonstrated consistent, safe, and highly relevant responses.
Strength Areas:
• Excellent clarity and explanation of quality.
• High accuracy in QA fundamentals and SQL queries.
• Strong consistency across similar prompts.
• No safety issues identified.
Areas for Improvement:
• Scenario-based responses sometimes lack negative test cases.
• Edge-case handling could include deeper coverage.
• Completeness varies for advanced reasoning tasks.

5. SAMPLE FINDINGS
Example 1:
Prompt: Difference between smoke and sanity testing Result:
Accurate and clear explanation; strong completeness.
Example 2:
Prompt: Boundary tests for DOB (1960–2005)
Result: Accurate but could include additional invalid edge values.
Example 3:
Prompt: SQL query for duplicate email detection
Result: Fully accurate and complete with proper use of GROUP BY and HAVING.

6. CONCLUSION
This project demonstrates ChatGPT's effectiveness as a QA learning and knowledge-support tool. The evaluation shows that ChatGPT provides highly accurate, relevant, and clear responses for most QA-related queries.
However, the completeness of responses in scenario-based and advanced QA prompts requires human review. Overall, this evaluation reflects strong analytical, test design, and AI assessment skills—key competencies for AI-driven QA and technical program roles.